

Non-Infectious Disease: Module 8

Homeostasis:

- Homeostasis:
- Is the maintenance of a relatively constant internal state by an organism
 - Regardless of changes in external environment
- Homeostasis allows organisms to optimise metabolic efficiency
- By allowing enzymes to function effectively
- Conditions being regulated:
- Blood pressure
- Temperature
- pH
- Water
- Concentrations of salt
- Glucose
- Oxygen
- Carbon dioxide
- Metabolic wastes

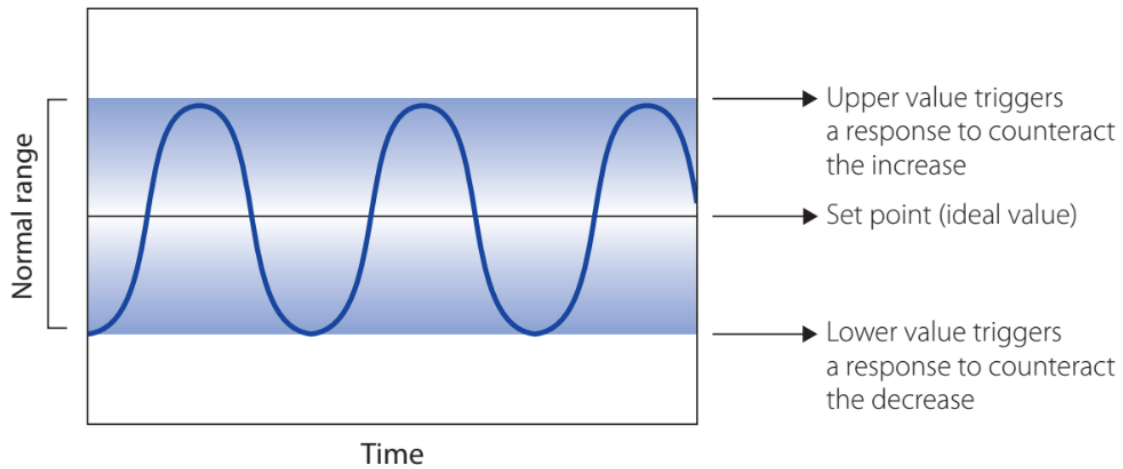
Importance:

- Metabolic efficiency:
- All chemical reactions in cells must occur efficiently
- + be effectively coordinated
- Reactions catalysed by enzymes:
- Enzymes sensitive to changes in environment
- $\therefore$ internal conditions maintained at level where
 - Enzymes can function optimally
 - $\therefore$ Optimal metabolic efficiency

Maintenance of Homeostasis:

- Involved in homeostasis:
- Nervous system
- Endocrine (hormonal) system
- Variables in internal environment are maintained within narrow range
- Narrow range aka Tolerance limits
- Each variable has ideal or normal value:

- Set point



- Homeostasis maintained when:
 - Minor fluctuations in tolerance limit around set point
 - Major fluctuations in variables in external environment
 - Fluctuations go past upper or lower tolerance limit
 - Triggers mechanism to bring body to normal set range

Negative Feedback System:

- Homeostasis: 2 main stages:

1. Detecting change:

- Sensory cells or receptors:
 - Detect change in specific component of internal environment
 - Change:
 - Stimulus

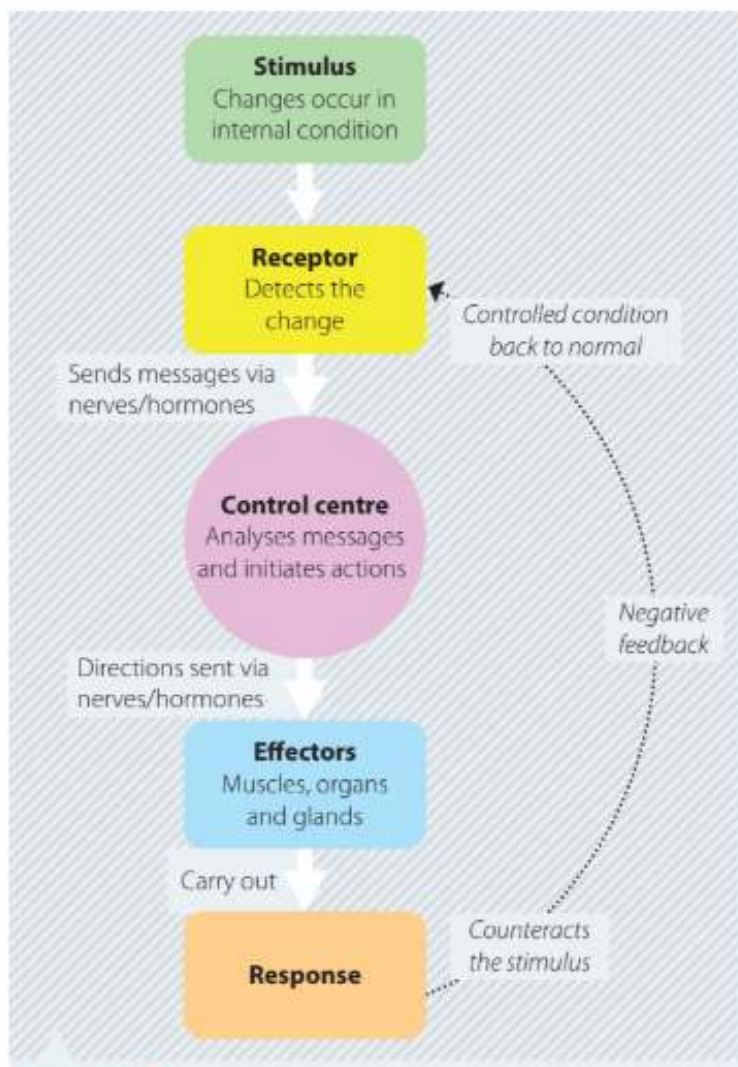
2. Counteracting Change:

- Response occurs that will reverse or counteract the change
- Response carried out effector organs
- Will restore body to relatively constant internal state

- Control centre:

- Link between 2 stages
- Maintains fluctuations around set point
- Receives information from receptors about change in condition
 - Which is:
 - Too above
 - Or too below
 - Set point
 - Decides appropriate response
 - Sends message to effectors

- to carry out actions that:
 - Make response reverse to counteract stimulus
 - Returns levels to set point
- When acceptable levels achieved
- Detected by receptors
- Receptors → control centre → effectors
 - Stops actions
- “Negative” as:
 - Response reverses stimulus
 - Correcting the change and returning it to set point



Coordination of Homeostatic Mechanism:

- negative feedback systems:
- Coordinated by:
 - nervous system
 - Endocrine system
 - Or combination of both

- Depends on what condition in internal environment involved
- **Nervous system** communicates in body by:
- Relaying information as:
 - **Nervous impulses**
- **Endocrine system:**
- Relaying information through:
 - Release of chemical messengers:
 - **Hormones**
- Hypothalamus:
- Region in lower central part of brain
- Control centre for maintaining most variables of internal homeostasis
- Has receptors for:
 - Body temp
- Acts as link between nervous + endocrine system
- Sends messages to effectors to:
 - Carry out responses for
 - Maintenance of many homeostatic conditions

Negative Feedback Loops:

- Regulated by **nervous system**

Body Temperature Regulation:

- **Thermoreceptors:**
- Detect change in stimulus
- **Hypothalamus:**
- **Control centre** of body temp regulation

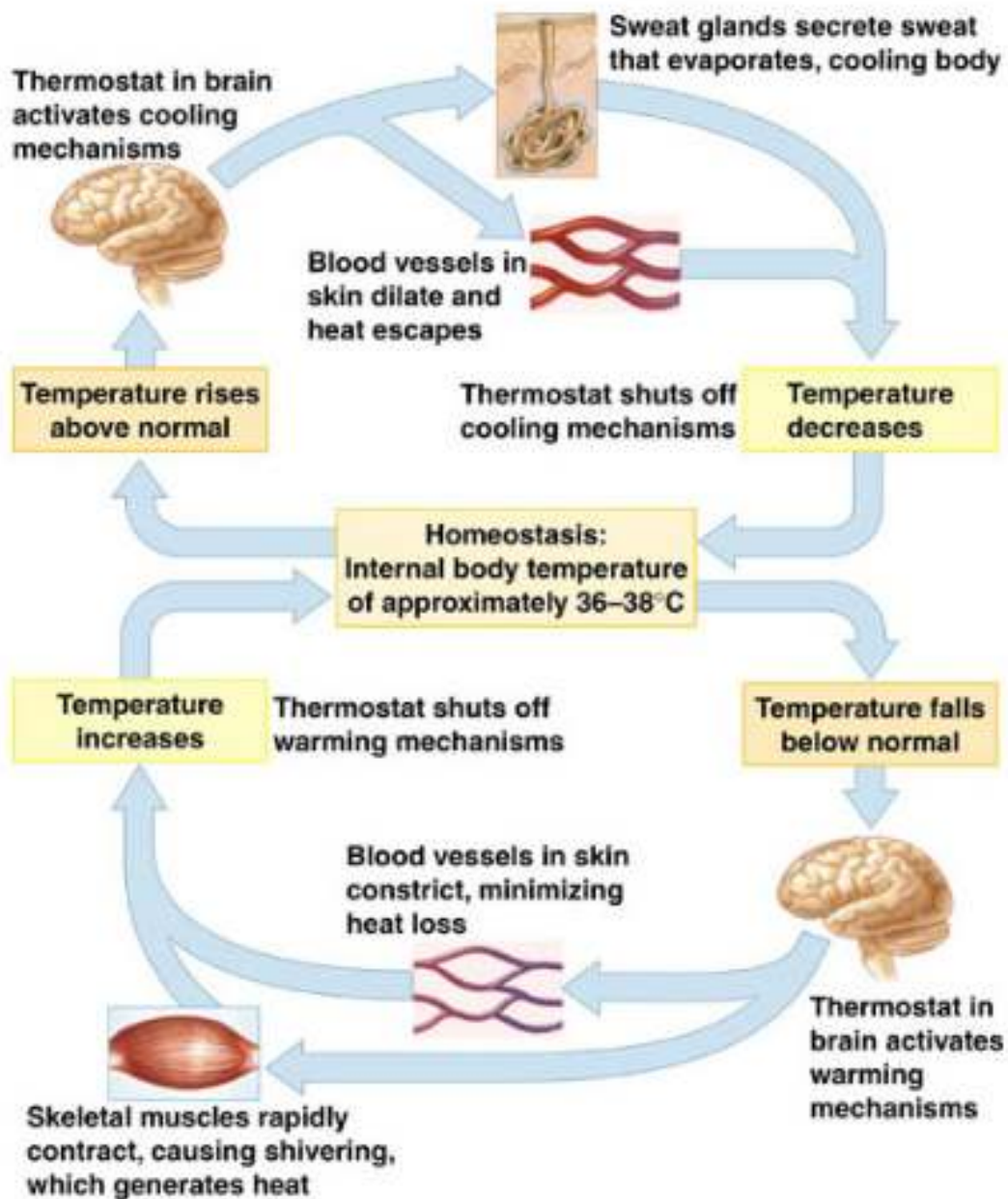
Cooling body:

- Increase in body temp
- Messages sent to front area of hypothalamus
- Processes to lose heat + keep body cool:
- **Vasodilation:**
 - Blood vessels dilating (expanding)
 - Brings blood closer to skin
 - **Heat escapes**
- Activation of **sweat glands:**
 - To secrete liquid sweat
 - Removes heat from body when it evaporates

- **Thyroid gland:**
 - **Lower rate of metabolism in cells** by
 - **Reducing amount of thyroxin produced**

Warming body:

- Decrease in body temp
- Messages sent to back area of hypothalamus
- Processes to conserve heat + warm body:
 - **Vasoconstriction:**
 - Blood vessels constricting (narrowing)
 - Removing blood from skin surface
 - Heat conserved
 - **Hair stands on end:**
 - **Traps layer of warm air** around body
 - **Release of thyroid-stimulating hormone**
 - By pituitary gland
 - Causes **increase in release of thyroxin**
 - **Increase rate of metabolism**
- **shivering**
 - Rapid contraction of muscles



Blood Glucose Regulation:

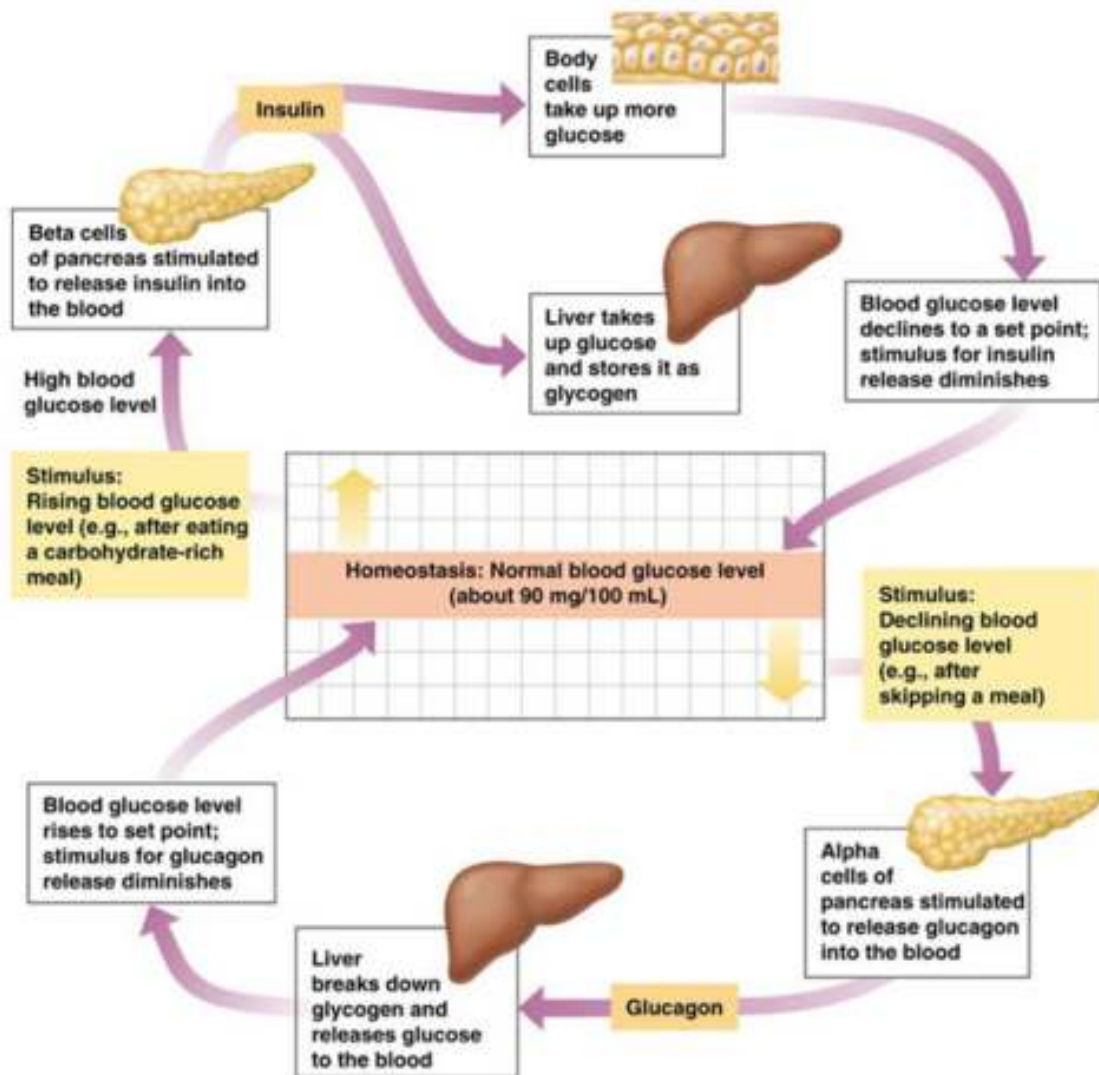
Increase in Blood Glucose:

- Stimulus:
 - Increase in blood glucose
- Receptor:
 - **Beta cells**
 - In pancreas
 - Release insulin
 - Remember BI (beta increase)
 - I Insulin for INcrease
- Control centre:

- Pancreas
- Effector:
- Liver:
 - Takes in glucose
 - Stores it as glycogen
- Body cells:
 - Take in more glucose
- Response:
- Blood glucose level goes down to normal
- Stimulus for insulin release finishes

Decrease in Blood Glucose:

- Stimulus:
- Decrease in blood glucose
- Receptor:
- Alpha cells:
 - In pancreas
 - Release glucagon
- Control centre:
- Pancreas
- Effector:
- Liver:
 - Breaks down glycogen → glucose
 - Release glucose into blood
- Response:
- Blood glucose rises to normal
- Stimulus for glucagon release finishes



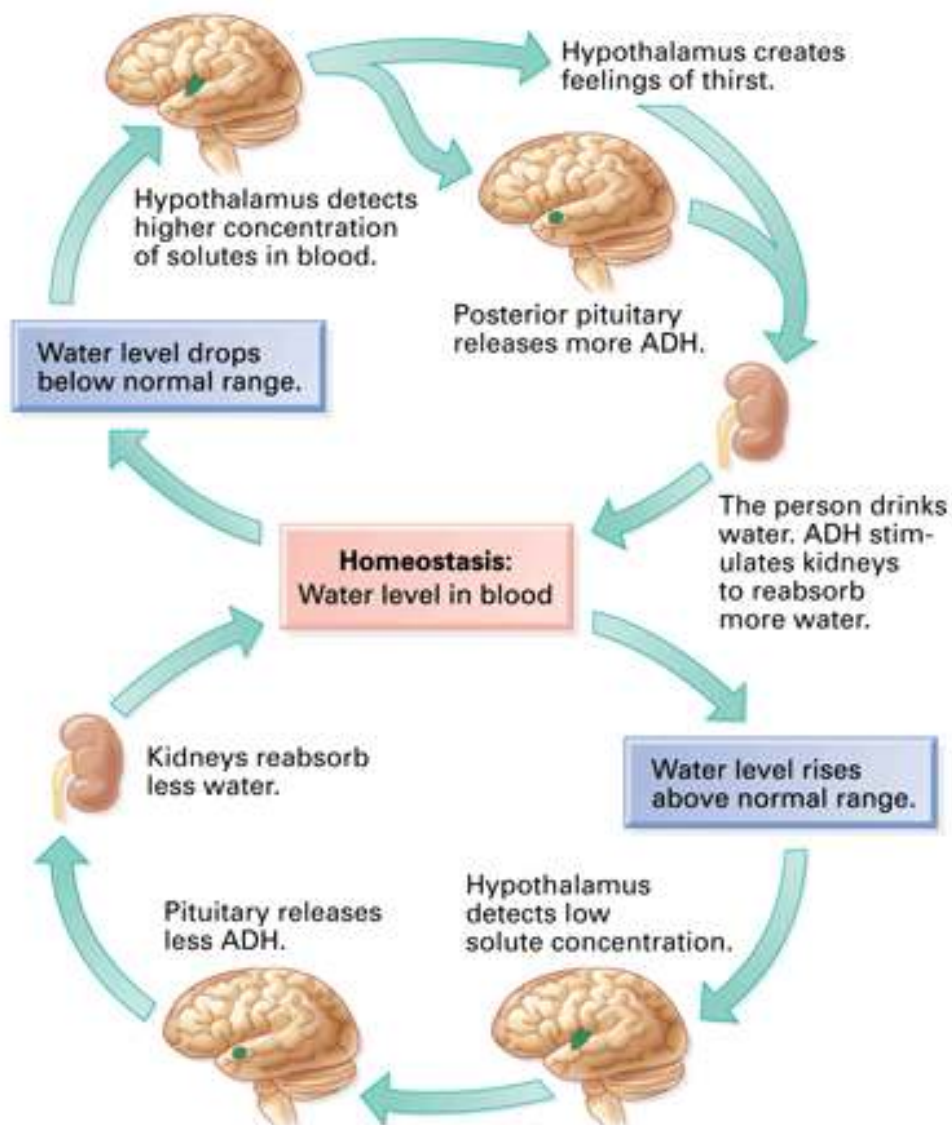
Water Level Regulation:

Increase in Water Level:

- Stimulus:
 - High water level
- Receptor:
 - **Osmoreceptors** in hypothalamus
- Control centre:
 - **Hypothalamus**
- Effector:
 - pituitary gland release less ADH into blood
 - ADH Causes kidneys to reabsorb more water
 - $\therefore$ less release $\rightarrow$ kidney absorb less water
 - Bladder release more urine
- Response:
 - Water levels decrease $\rightarrow$ normal

Decrease in Water Level:

- Stimulus:
 - Low water level
- Receptor:
 - **Osmoreceptors** in hypothalamus
- Control centre:
 - **Hypothalamus**
- Effector:
 - **pituitary gland increase ADH** in blood
 - **Hypothalamus creates feelings of thirst**
 - **Bladder release less urine because of conserving water for body**
- Response:
 - Person drink more water
 - Increase water level



Mechanisms to Maintain Homeostasis:

- Nervous + endocrine systems
- Work together to:
 - maintain homeostasis
 - Make pathway for relaying messages through:
 - Nerves or
 - hormones
- Role of these systems:
- Coordinate
- Provide communication pathways for
 - Negative feedback systems
 - To maintain homeostasis
- Nervous system → nerve impulses:
 - Transported quickly along nerves to body locations
- endocrine system → hormones:
 - Chemical substances
 - Transferred to body regions in bloodstream
 - At much slower rate

Internal Coordination Systems:

Receptors:

- Have sensory cells
- Complex form receptors:
- Concentrated in specific area
- Make sense organs
 - E.g. eyes, ear, tongue
- Receptors in sense organs detect stimuli
- in external environment
- Interoceptors:
- Receptors in body that
 - Detect internal stimuli related to homeostasis

Receptors in Homeostasis:

- Thermoreceptors:
- Detect changes in temperature
- In skin are:
 - Nerve endings

- Sensitive to heat + cold
- Send information to brain about external temp
- Internal thermoreceptors in hypothalamus:
 - Detect temp of blood
 - When it flows through brain
- Chemoreceptors:
 - Detect concentration of chemicals in body
 - Found in
 - Certain blood vessels
 - Detect:
 - pH
 - Levels of certain chemicals
 - Co2
 - O2
- Osmoreceptors:
 - Detect changes in osmotic pressure
 - Found in
 - Hypothalamus
- osmotic pressure in blood determined by:
 - Concentration of substances dissolved in blood plasma

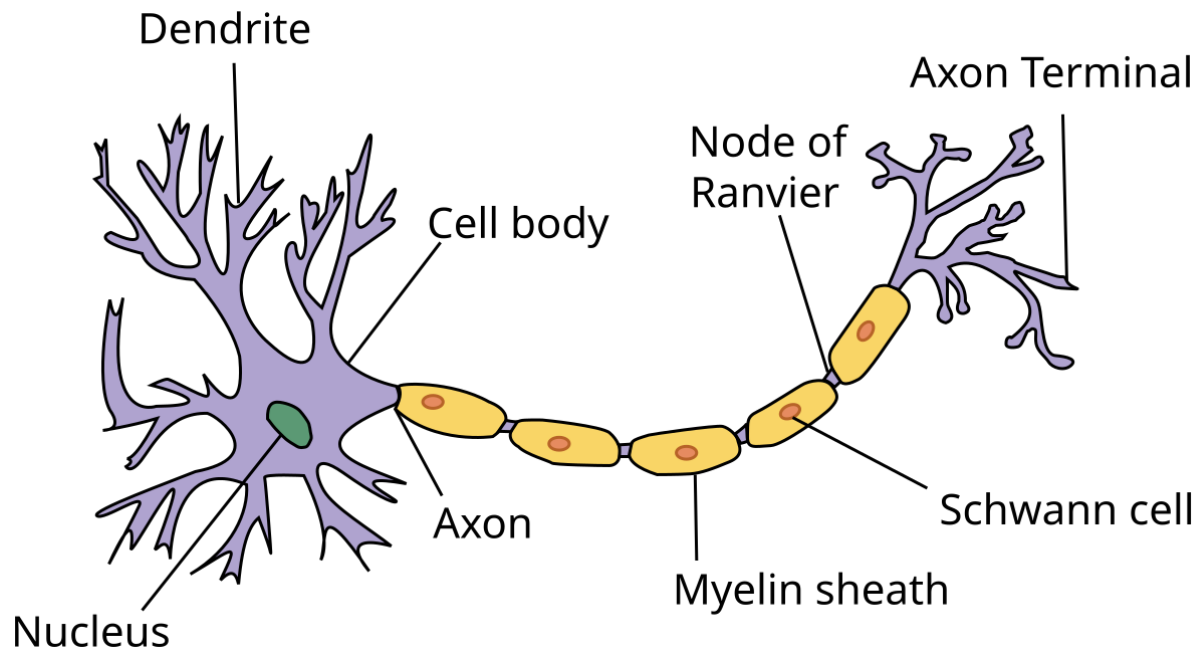
Nervous System:

- nervous system
- Gives neural pathways
- Acts as control centre
- 2 parts:
 - Central nervous system (CNS):
 - Brain
 - Spinal cord
 - Peripheral nervous system (PNS):
 - Includes all other nerves over body that are not part of CNS
 - Carry information to and from CNS
 - Messages carried → Electrical impulses

Neurons:

- Nerve cells or neurons
- Structure:
- Cell body:

- Nucleus
- Many organelles
- Nucleus + organelles form grey matter of CNS
- 1+ fine branching extensions:
 - Dendrites
 - Extensions of cytoplasm + body
 - Get messages in form of impulses from
 - Other axons
 - Send impulses → cell body
 - Sensory neurons
 - Dendrite aka dendron
- Axon:
 - 1 single long extension of cytoplasm of cell body
 - Send messages away from cell body
 - Form white matter of CNS
- Types of neurons:
- Sensory neurons:
 - Carry impulses from sensory cells → CNS
- Motor neurons:
 - Transfer messages from CNS → effectors
 - Effectors e.g.: muscles, glands
- Interneurons:
 - Link between sensory and motor neurons
- When nervous impulse transferred from axon → dendrite
- Cross gap or synapse
- Fibres of neurons
- Gathered into bundles held by
 - Connective tissue sheath
 - Gives structures pathway for transmitting electrical impulse



Central Nervous System:

- Made Of 2 types of tissue:
 - Grey matter
 - Neuron cell bodies
 - White matter
 - Nerve fibres surrounded by myelin sheaths

Brain:

- Main control centre of body
- Controls maintenance of homeostasis

Hypothalamus:

- Located centrally
- Close to pituitary gland
- Control centre for regulation of homeostasis
- Basically everything homeostasis control by hypothalamus
 - Except blood glucose
 - Blood glucose cc is pancreas
- Regulates by:
 - Telling effectors to carry out response by send messages via
 - neural pathways
 - Chemical messages (hormones)
- Link between nervous + endocrine system
- directs hormone secretions

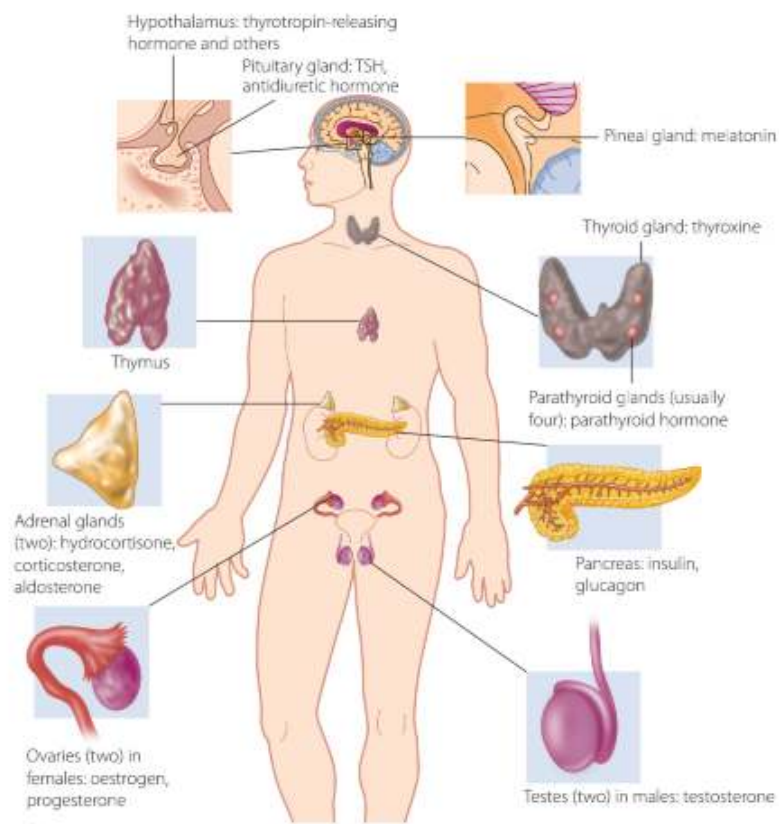
- Directs actions of pituitary gland in coordinating other glands
 - To secrete hormones
 - To maintain homeostasis
- Functions:
 - Main function to keep body at stable state - homeostasis
 - Other functions:
 - Release hormones → pituitary gland → control other glands
 - Maintaining other cycles like
 - Sleep
 - Heart rate
 - Breathing
 - Managing sexual behaviour
 - Regulating emotions

Spinal Cord:

- Contains nerve fibres that
- Give link for pathway of nerve impulses
 - Between peripheral nervous system + brain
- 2 main functions:
 - Act as conduction pathway for nerve impulses
 - From receptors around body → brain
 - For nerve impulses from brain → effectors
 - Important for all area of body to function
- Coordinates reflex actions

Endocrine System:

- Hormones:
 - Chemical messengers
 - Secreted by endocrine glands
- Transported by blood → cells
 - To change their activity in a way that
 - Maintains homeostasis
- Hormones change cell activity by:
 - Influencing activity of particular enzymes
 - Or changing conc. of enzymes in cells



- Major Glands in body



Pituitary Gland

- MASTER GLAND

- Works closely with hypothalamus
- Releases hormones in direction of hypothalamus
- To regulate activity of other glands
- Front:
 - Controlled by hormones released by hypothalamus
- Back:
 - Controlled by nerve impulses
- Hormones:
 - Hormones secreted by front:
 - Controls growth
 - Act on other glands to control activity of:
 - Thyroid
 - Adrenal gland
 - Gonads (ovaries + testes)
- ADH (Antidiuretic hormone)
 - Helps regulate conc of water in body

- If receptor cells in hypothalamus detect:
 - Water too low:
 - Hypothalamus stimulates pituitary gland to release ADH
 - Acts to conserve water in body by promoting its reabsorption by kidney tubules
 - Water too high:
 - Hypothalamus directs pituitary gland to reduce ADH
 - Leads to less water being absorbed by kidneys
 - Also increased secretion of water

Thyroid Gland:

- Functions:
 - Regulate metabolism
 - Control energy production
 - Promotes growth + development of children
- Produce thyroxine
- Production controlled by thyroid stimulating hormone
 - produced by front pituitary gland
- Thyroxine:
 - Tells cells to increase their metabolic rate
- Parathyroid gland:
 - 4 small glands
 - On surface of thyroid gland
 - Maintains calcium level in blood
 - By secreting parathyroid hormone

Adrenal Glands:

- triangular
- Located on top of each kidney
- Produces hormones that regulate:
 - Metabolism
 - Blood pressure
 - Immune response
 - Hypothalamus regulates activities
- Produces adrenaline - stress response:
 - Main hormone produced
 - Responsible for fight or flight response
- Causes heart to race, breath to quicken, increases blood flow to muscles and brain

- Response help us fight or run away from danger

Thymus:

- Major role in protecting body from
- Pathogenic threats
- Does this by secreting hormone
- Thymosin
 - Stimulates production of white blood cells (T cells)

Pancreas:

- Produces enzymes that
- Break down:
 - Sugars
 - Fats starches
 - In food
- Hormones produced:
 - Insulin
 - glucagon

Testicles:

- Produce testosterone

Ovaries:

- Produce hormones:
- Oestrogen
- Progesterone
 - Both play role in:
 - Sexual function
 - Regulating menstrual cycle
 - Ovulation

Adaptations in Endotherms:

- Endotherms:
- Organisms able to maintain body temp
- Despite external environment temp changes
- Adaptations:
- Behavioural:
 - The way an organism acts

- Structural:
 - Physical characteristics of organism
- Physiological:
 - The way an organism's body functions

Behavioural Adaptation:

- Example:
- Changing position of body to be
 - more or less exposed to sunlight
- Nocturnal activity:
 - Being passive during day
 - Being active at night
- Migration:
 - Organisms moving to habitat
 - That is close to their ideal temp

Structural Adaptation:

- Example:
- Insulation:
 - Using hair, fur, feathers
- SA:V ratio:
 - Large SA: V:
 - Lose heat
 - Small SA:V:
 - Keep heat

Physiological Adaptations:

- Example:
- Changing metabolic activity
- Hibernation:
 - Long period of inactivity
 - Response to cold
- Torpor:
 - Decreasing metabolism, heart rate, respiratory rate
 - Response to heat
- Regulating blood flow

Mechanisms to Maintain Water Balance in Plants:

[Halophyte Adaptations](#) | [Elucidate Education](#)

- Transpiration:
- Evaporation of water from stomata
- Triggers continuous transpiration stream
 - Cools plant

Decreasing Internal Temperature:

- Leaves of plants:
- Covered in shiny waxy cuticle
- Or thick leathery cuticle
 - Prevents water loss
 - Cool plant
- White hairs:
 - Reflect sunlight
 - Reduce surface temp
 - Reduce water evaporation

Reducing Exposure to Sunlight:

- Leaf position
- Stomata not exposed to sunlight
- Reduced surface area of organs
- That have most amount of stomata
- Getting rid of transpiring organs
- Stop water loss through evaporation
- Need to have more adaptations to stop overheating, increase photosynthetic tissue or ensure pollination
 - Because of lack of leaf or petal sa

Opening + Closing of Stomata:

- Only opening stomata in cooler parts of day
- Closing stomata in hot parts of day
- Creating moist environment around leaves through:
- Hairy leaves
- Rolled leaves
- Sunken stomata
 - To keep stomata open for greater amount of time for photosynthesis

Water Storage:

- Succulents:
- Fleshy stems or leaves
- Swell to retain moisture

Fruits:

- Produce woody fruits
- Not fleshy fruits
- Reduce amount of water lost when fruits fall off

Non-Infectious Diseases:

Causes and Effects:

- Aka non communicable
- Not caused by pathogen
- Not contagious
- Risk factors:
 - Age
 - Gender
 - Economic and social conditions
 - Culture
 - Race
- Lifestyle
- Environment
- Genetics
- Nutrition

Genetic Diseases:

- Caused by mutations of genes or chromosomes of an individual
- Each gene:
 - Codes for production of specific protein
- Protein May be:
 - Structural
 - Enzyme
 - Hormone
- If gene that codes for specific protein abnormal:
- Protein not produced correctly
- Results in genetic disease

- May happen when:
- Cell division by mitosis or meiosis
 - Occurs abnormally
 - Results in cells with:
 - Wrong number of chromosomes
 - Abnormal chromosomes

Single Gene Abnormalities:

- Many **Single gene diseases caused:**
- **When person inherits mutated genes from parents**
- Abnormal recessive allele
- Needed from both parents for disease
- Abnormal dominant allele
- Needed from one parent for disease
- Is abnormal gene on sex chromosome
- Disease → sex linked
- If enzymes coded for by mutated gene
- Enzyme not made correctly
- Metabolic pathway not proceeding as it should
 - Results in genetic disease
- Examples of gene mutated diseases:
- Cystic fibrosis
- Albinism
- Huntington's chorea
- Thalassaemia
- Tay Sachs disease
- Sickle cell anaemia
- haemophilia

Cystic Fibrosis:

- Inherited disease
- Caused by mutation of CFTR (cystic fibrosis transmembrane conductance regulator)
- On **chromosome 7**
- **Changes protein that regulates movement of sodium chloride (salt) in and out of cells**
- Causes mucus secreting organs to
- Secrete abnormally thick mucus
- **Autosomal recessive**
- Person must have 2 copies of abnormal gene to have disease

- One gene from each parent
- Affects:
 - Respiratory
 - Digestive
 - Reproductive systems
 - Sweat glands
 - Mucus secreted is very thick
 - Blocks passageways in lungs, digestive tracts
- Causes problems with:
 - Breathing
 - Digestion
 - Absorption of nutrients
- Symptoms:
 - Recurrent + severe chest infections
 - Malabsorption of nutrients
 - Failure to thrive
 - Passing of large, foul smelling motions
 - Diabetes
 - Excess salt in sweat
 - Liver failure
 - Infertility
 - Average life expectancy: 30-40 years

Phenylketonuria (PKU):

- Autosomal recessive
- Caused by mutation
 - on gene chromosome 12
- That codes for enzyme phenylalanine hydroxylase
 - Required at 1st step of breakdown of
 - Amino acid phenylalanine
- PAH enzyme not made correctly
- Phenylalanine not broken down
- Leads to
 - Buildup of phenylalanine in blood + tissues
- Results in:
 - Development of intellectual disabilities
 - Delays in development
 - Social, emotional, behavioural problems

- Psychiatric disorders
- Hyperactivity
- Risk of seizures

Chromosomal Abnormalities:

- Genetic diseases may be caused by mutations of chromosomes
 - that carry genes
- Mutations:
 - Wrong number of chromosomes
 - Too many
 - Not enough
 - Change in chromosome itself:
 - Deletion
 - Addition
 - Altered sections of chromosome
- Non disjunction:
 - Wrong separation of chromosomes during cell division
 - End result:
 - Wrong number of chromosomes in zygote
- Trisomy:
 - One extra chromosome + normal chromosome pair
 - Total number of chromosomes in zygote
 - 1 more than normal diploid number of chromosomes
- Monosomy:
 - 1 member of chromosome pair missing
 - Total number of chromosomes in zygote
 - One less than normal diploid number of chromosomes
- Syndrome:
 - Group of symptoms occurring together
 - Characterise a particular disease

TABLE 15.2 Trisomy and monosomy syndromes and the chromosome involved in each

SYNDROME	CAUSE
Patau	Trisomy 13
Edwards	Trisomy 18
Down	Trisomy 21
Triple X	Trisomy X (XXX)
Klinefelter	XXY
Jacob	XYY
Turner	Monosomy X (XO)

- **Translocation:**

- Chromosomes mutated when
 - Section of 1 chromosome moves to join another
 - In different homologous pair
- No genes lost
- May alter phenotype expressed

Down Syndrome:

- **TRISOMY ON CHROMOSOME 21**

- Effects:
- Characteristic facial features:
 - Small flattened skull
 - Short flat bridged nose
 - Almond shaped eyes
 - Skin folds around eyes
- Protruding tongue
- Small ears folding over a bit at top
- Intellectual disability
- Short stature
- Heart defects
- Sexual underdevelopment + infertility
- Short stubby hands
- Looseness in joints
- Susceptibility to infection

Klinefelter Syndrome:

- Trisomy on sex chromosomes
- XXY
- When supposed be XY
- Effects:
 - Lower levels of testosterone
 - Unusually small testes - sterile
 - Taller than average height
 - Some have
 - Degree of breast enlargement
 - Feminine fat distribution on
 - Abdomen
 - Hips
- Behavioural, learning problems

Diseases Caused by Environmental exposure:

Lifestyle Diseases:

- Caused as direct result of how people live their lives
- Most common:
 - Cardiovascular disease
 - Diabetes
 - Cancers
 - Chronic lung disease
- Risk factors:
 - Tobacco use
 - Unhealthy diet
 - Physical inactivity
 - Harmful use of alcohol
- Exposure to additives and hormones

Atherosclerosis:

- "Hardening of arteries"
- Factors:
 - Insufficient physical exercise
 - Excessive drinking of alcohol
 - Exposure to high levels of stress
 - Smoking

- Unbalanced diet too high in:
 - Kilojoules
 - Saturated fats
 - Salt
 - Red meat
- Lifestyle activities followed for long time:
- Leads to:
 - Deposition of liquids (fatty deposits)
 - In inner walls of arteries
- Continued deposition
- Internal walls of arteries:
 - Not smooth
 - Very rough + thickened
- Stops blood flow
- Increases blood pressure (hypertension)
- Plaque buildup on artery walls
- Reduces elasticity + blood flow
- Increases hypertension
 - May cause cerebral hemorrhage
 - Leakage of blood into brain tissue
- Blood clots may form + build up of fat + plaque
- Cause blockage in blood vessel
 - Can be total or partial
- Blockage in arteries:
- Cardiac arrest
- Heart failure
- Blockage in brain:
- stroke

Physical Factors:

- UV light from sun or in solariums
- Radiation from nuclear substances
- Factors affect health of people by:
 - Changing the genetic material in some way
 - Causing problems with correct functioning of body
- Excessive exposure to UV light:
 - Skin cancer
 - UV light:

- Changes DNA of skin cells
- May cause continued abnormal cell division

Exposure to Chemicals:

- Type of disease + period of exposure necessary for disease:
- Varies depending on chemical
- Examples:
- Toxic discharges into atmosphere or waterways
 - Result of industrial processes
- Chemical exposure in workplace
- Pollution of air
- asbestos

Asbestosis:

- Caused when inhaled asbestos fibres
- Cause inflammatory reaction in lung tissue
- Inflammation leads to:
- Scarring + stiffening of lung tissue
 - Making breathing harder
 - Reduces amount of oxygen taken in by blood
- Symptoms:
- Shortness of breath
- Persistent cough
- Chest pain
- Appetite loss
- Enlarged fingertips

Nutritional Diseases:

- Caused by diets lacking proper balance and amount of nutrients
- May be caused by psychological conditions that
- Lead to inappropriate diets
- Malnutrition:
- Imbalance in diet
- 2 categories:
 - Undernutrition:
 - Insufficient intake of correct type of food
 - Quantity of food
 - Combination of both

- Overnutrition:
 - Excessive intake of food

Undernutrition:

Lack of Protein + Energy Rich Food:

- Kwashiorkor
 - In countries where food often in short supply
 - Children especially
 - Caused by severe lack of protein in diet
 - Results in:
 - failure to grow
 - Enlarged liver
 - Hair changes
 - Apathy
 - Irritability
 - Increased susceptibility to infectious diseases
 - Characterised by:
 - Swollen belly
 - Caused by fluid retention

Lack of Vitamins

- Vitamin A, D, B1 deficiencies

Lack of Minerals

- Iron deficiency - anaemia
- Iodine deficiency

Eating Disorders:

- Have a complex psychological basis
- Anorexia nervosa
- Fear of putting on weight
- Bulimia nervosa
- People don't become as thin

Overnutrition:

- Obesity:
 - Caused by consuming more kilojoules than energy expended
 - Results in accumulation of fat in body

- Effects:
 - Increased blood pressure
 - Atherosclerosis
 - Gallbladder disease
 - Stroke
 - Type 2 diabetes
 - Problems With weight bearing joints
 - Increased risk of developing certain cancers:
 - Colorectal cancers
 - Reduced life expectancy

Cancer:

- Disease of cells of body
- Happens when:
 - Abnormal cells divide in an uncontrolled way
 - Cells can invade body tissue near them
 - Spread to other tissues
 - Disrupt normal functioning of body
- Cells division:
 - Controlled by specific set of genes
 - In DNA of cell
- 3 Types of genes that control correct functioning of cell division:
 - DNA repair genes:
 - Fix errors in DNA replication
 - Proto-oncogenes:
 - Code for proteins that
 - Stimulate cell growth + mitosis
 - Tumour suppressor genes:
 - Code for proteins that:
 - Slow down
 - Stop cell growth + mitosis
 - Induce cell death if uncontrolled increase in cell numbers
- Often all 3 types of genes need to be damaged for
- Cells to become cancerous
- Disruption of normal cell cycle:
 - Cells not differentiating
 - Forms tumours:
 - If tumour is invasive or spreading

- Cancer results
- Breast Cancer:
 - BRCA 1:
 - PTEN gene (Tumour suppressor gene)
 - Chromosome 17
 - Responsible for coding proteins
 - For repair of PTEN gene
 - Limits amount of cells division
 - Encourages cell death
 - Regulates cell cycle
 - Limits replication of cells
 - That lead to tumours, cancers
- Mutations in BRCA 1:
 - Put women at higher risk of developing breast cancer
 - Results in:
 - Non-production of proteins necessary for
 - Repairing damage of PTEN gene
 - PTEN gene then stays damaged
 - Leads to:
 - Lack of control of cell cycle
 - + runaway cell division
 - ∴ formation of tumours

Tumours:

- Primary tumour:
 - Area in body where first malignant tumour develops
- Secondary tumour:
 - Tumours moved to other parts of body
- Benign tumours:
 - Not cancer
 - Don't spread
- Malignant Tumours:
 - Have cancerous cells
 - Replicate quickly in disorganised manner
 - Direct nutrients to themselves, away from tissue next to them
 - Invade surrounding tissue
 - Some abnormal cells break away from tumour
 - Enter blood or lymphatic system

- Travel to new part of body
 - Develop into more tumours
- Metastasis
 - Process of spreading

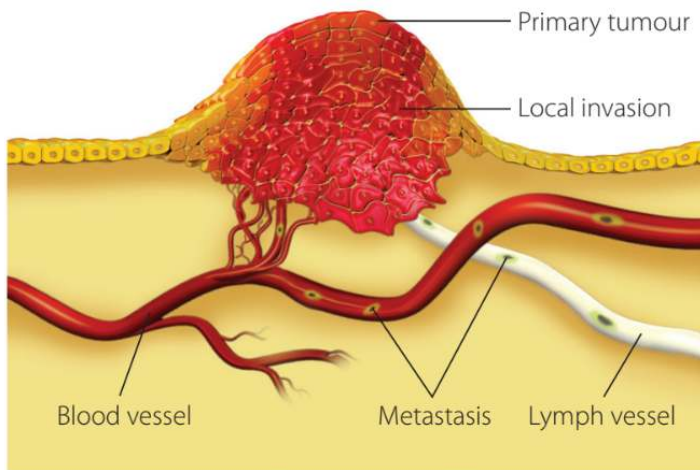


FIGURE 15.20 The process of metastasis – how cancer spreads. Angiogenesis, the development of new blood vessels, provides the abnormal cancerous cells with nutrients.

Causes:

- Agents aka carcinogens:

- Chemical
- Physical
- Biological
- Risk factors:
 - Smoking
 - Excessive alcohol consumption
 - Lack of physical activity
 - Being overweight, obese
 - Having diet high in saturated fats
 - Exposure to radiation
 - UV radiation
 - Getting viruses
 - Hpv
 - Hepatitis b and c
 - Exposure to certain chemicals
 - Inheriting mutated genes

Melanoma:

- Skin cancer
- Melanocytes with change in DNA

- Divide uncontrollably

- Risk factors:
- UV light exposure
- Fair complexion
- Blue or green eyes
- Blonde or red hair
- Skin that burns easily
- Severe sunburn causing blistering in childhood
- Having history of skin cancer
- Having many moles or irregular looking moles
- Family history of moles, melanoma
- Exposure to certain environmental conditions or chemicals
 - Radiation
 - Organic solvents
- Compromised immune system
- Older age
- Male gender

Effect:

- Tumour left untreated
- spreads into deeper layers of skin
- More time:
- Spreads to:
 - lymph nodes
 - Lymph vessels
 - Skin in areas
 - Other organs
- Interferes with body normal function
- Death

Incidence, Prevalence and Mortality Rates:

- Incidence:

- Amount of new cases of disease

- Reported in specific time period

- (normally a year)

- Prevalence:

- Number of people in a specific population

- That have been diagnosed with disease ever

- Are still alive at end of given time period
- Mortality rate:
- Number of deaths due to a particular disease
- In specific time period
- (normally 1 year)
- Normally expressed as a number per 100 000 population

Epidemiology:

- Scientific study of patterns of
- occurrence of disease in human populations
- Statistical analysis → formed for cause of disease in population
- Results used:
- To develop strategies to
 - control disease
 - Improve public health
- Evaluate strategies already in place

Epidemiological Studies:

Descriptive Studies:

- Purpose:
- To IDENTIFY + DESCRIBE DISEASE PATTERNS in populations
- Find commonalities with those affected → potential causes
- Data gathered:
- Age
- Sex
- Occupation
- Etc
- Provide information about:
- Disease patterns
- Frequency of disease
- Demographics affected
 - Age
 - Sex
 - Occupation
- Analyse:
- Geographic location
- Specific time periods linked to disease

- BASICALLY:
- People with disease, their demographics
- Compared to people without disease, their demographics
- Find common causes in many people affected
- Describing trends + patterns of disease

Analytical Studies:

- After descriptive study
- More data collected + analysed to get hypothesis about cause of disease
- Data may have:
- Morbidity:
 - Number of cases
- Mortality:
 - Number of deaths
- Incidence rates:
 - New cases in specific period
- Prevalence:
 - People affected at any one time

Case-Control:

- Comparing people with disease (cases)
- To without disease (control)
- Aim:
- To identify differences in exposure to potential causes of the disease
- Data collected on factors:
- Age
- Sex
- Diet
- Location
- Lifestyle
- occupation

Cohort:

- Method:
- Long term follow up (years usually) of exposure groups
 - To determine risk

Intervention Studies:

- Evaluates treatment and prevention strategies
- Through trials

Study Must Haves:

Studies should:

- Last long time
- Large sample size (1000's)
- Collect lot of data from affected + unaffected people
- Have participants that
 - represent large part of society
- Use controls that
 - Are not exposed to potential cause of disease
- Statistically analyse data for patterns and trends
- Identify possible cause of disease
- Develop management plan
- Evaluate effectiveness of control + treatment programs

Treatment, Management and Future Directions:

Melanoma:

- Disease caused by environmental exposure

Treatment:

- If detected early:
- Surgery
- To remove tumour + skin around it
- Advanced cases:
- Radiation
- Chemotherapy
- Targeted therapies
- Immunotherapy

Radiation:

- Exposing cancer cells to x rays or other radiation
- Kills them by
 - Damaging their DNA
- External or internal in body

Chemotherapy:

- Slow growth of cancer cells
- Not used for melanoma

Targeted therapies:

- Drugs → affect molecules controlling tumour cell growth

Immunotherapy:

- Checkpoint inhibitors:
- Cause immune system to recognise + destroy cancer cells
- Vaccinations:
- Antigen from cancer cells
- Allows immune system to easier identify + destroy cancer cells

Future Directions:

- Future research
- Unknown mutations causing uncontrolled cell division
- For more immunotherapy drugs
- Vaccines to cancer

Evaluating an Epidemiological Study:

- Basically like evaluating or making (depending on q) experiment

How to evaluate:

- Was aim of study achieved
- Hypothesis
- (...more likely to....because...)
- Was appropriate type of epidemiological study used
- Reliability:
- Validity:
- Done over long period of time
- Very large group
- Collect relevant data from affected + unaffected people
- (set questionnaires)
- Have people that represent wide range of society + lifestyle
- (no person eliminated because of their answers)
- Control group:
- Show independent variable → cause of change
- Data on:

- Incidence
- Prevalence
- Mortality
- Morbidity rates
- Identify possible cause of disease + risk factors
- Development of management plan with strats to control disease
- Evaluate the effectiveness of control + treatment programs
- Final data peer reviewed
- Describe what to do to analyse data

Benefits of Epidemiological Studies:

- Helps:
 - determine cause of disease
 - Who is affected from disease
 - Finding patterns to identify individuals at risk
- How to intervene to
 - Prevent further occurrence of disease
- Evaluate strategies already put in place to control + treat
 - Disease
 - Health related conditions
- Developing strategies for:
 - Controlling disease
 - Improving public health

Examples:

- Strategies for improving public health:
- Campaigns:
 - Lung cancer campaigns targets smokers
 - Melanoma campaigns sun safety
- Who is affected by diseases + how to intervene
- Women aged between 50-74 advised to regularly screen for breast cancer
- Diabetes:
 - Canteen selling healthier food
 - Compulsory PE til year 10
 - Alcohol + Smoking:
 - Healthy harold education program to influence children in developing healthy habits
 - Smoking:
 - No smoking on

- Public transport
- Public parks
- Workplaces

Prevention of Non-Infectious Diseases:

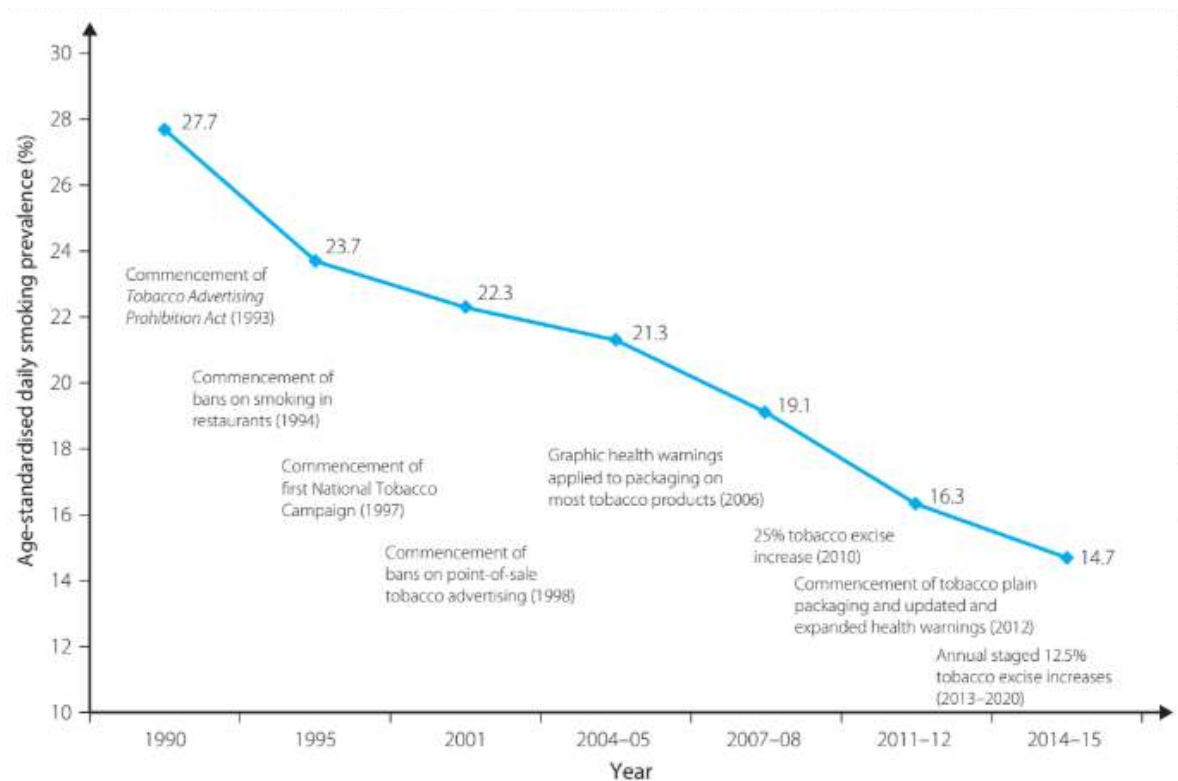
Educational Programs and Campaigns:

- Has:
- An evidence base for action
- A package of a limited number of high priority evidence-based interventions
- Effective performance management with:
 - Real-time:
 - Monitoring
 - Evaluation
 - Program management
- Partnerships between the public and private sectors
- Communication Of accurate information to healthcare workers + public
- Political commitment
- Examples:
 - “Slip, slop, slap, seek, slide”
 - QUIT program
 - Diabetes Helpline

QUIT Campaign:

- Smoking is leading preventable cause of death in Australia
- 15 500 smoking related deaths per year
- Epidemiological studies:
 - Showed clear link between cigarette smoking + reduced life expectancy
- National institute of health states that:
 - Risk of dying of lung cancer:
 - 23x higher in men
 - 13x higher in women
 - Than it is for non smokers
- Strategies:
- Slogan use e.g.:
 - Quit for life
 - Quit B fit
 - iCanQuit

- Disturbing graphic images in media + cigarette packaging
- National helpline
- Legislation:
 - Ban on all tobacco advertising, promotion, sponsorship
 - Ban on smoking in workplace + public places
 - Plain paper packaging of cigarettes having:
 - Pictorial, graphic warnings + no logos, colours, brand images or promotional information
 - Increase in taxes on tobacco products
 - Keeping cigarette packets out of sight of consumers



Advantages:

- Reaches a large population through multiple platforms
- No. of smoking population has decreased
- $\therefore$ no. of lung cancer incidence decreased

Disadvantages:

- Doesn't actively enforce change
- People will ignore information

Genetic Engineering:

- Using gene manipulation to prevent non-infectious diseases

Pre-implantation Genetic Testing:

- Starts with IVF (in vitro fertilisation)
- Eggs are harvested and collected
- Eggs fertilised by direct injection of sperm
- Embryos grown for 3 days
- One cell from 8 cell embryo removed
- Cell tested for mutated gene
- Embryos free of gene or were carriers retained
- Other embryos containing gene destroyed or used for research
- On day 5:
- One of retained embryos implanted into female uterus
- Rest of embryos frozen for later use

Advantages:

- More ethical:
- Than testing + aborting foetus
- Cost saving:
- Family + health system not having to pay for future treatment
- Carrier removed + discarded
- Removes disease from family generations in future
- High accuracy

Disadvantages:

- Some people culturally against IVF
- Discarding embryos is unethical

Preventing Nutritional Diseases - Golden Rice:

- Rice that includes gene from maize + soil bacterium
- Allows rice to produce beta-carotene
- Used by body to produce vitamin A
- To be used in africa and asia
- Deficiency in vitamin A:
- May lead to:
 - Blindness
 - Compromised immune system

Advantages:

- Rice is staple food in many countries

- Easily incorporated into daily diet
- Reduces risk of blindness
- Which is caused by lack of vitamin A
- Seeds easily replanted

Disadvantages:

- Smaller yield compared to normal rice
- Opposition against genetic modification due to culture, ethics, religions
- May have increased cost
- Not affordable for production in developing countries

Evaluating Effectiveness of Disease Prevention Methods:

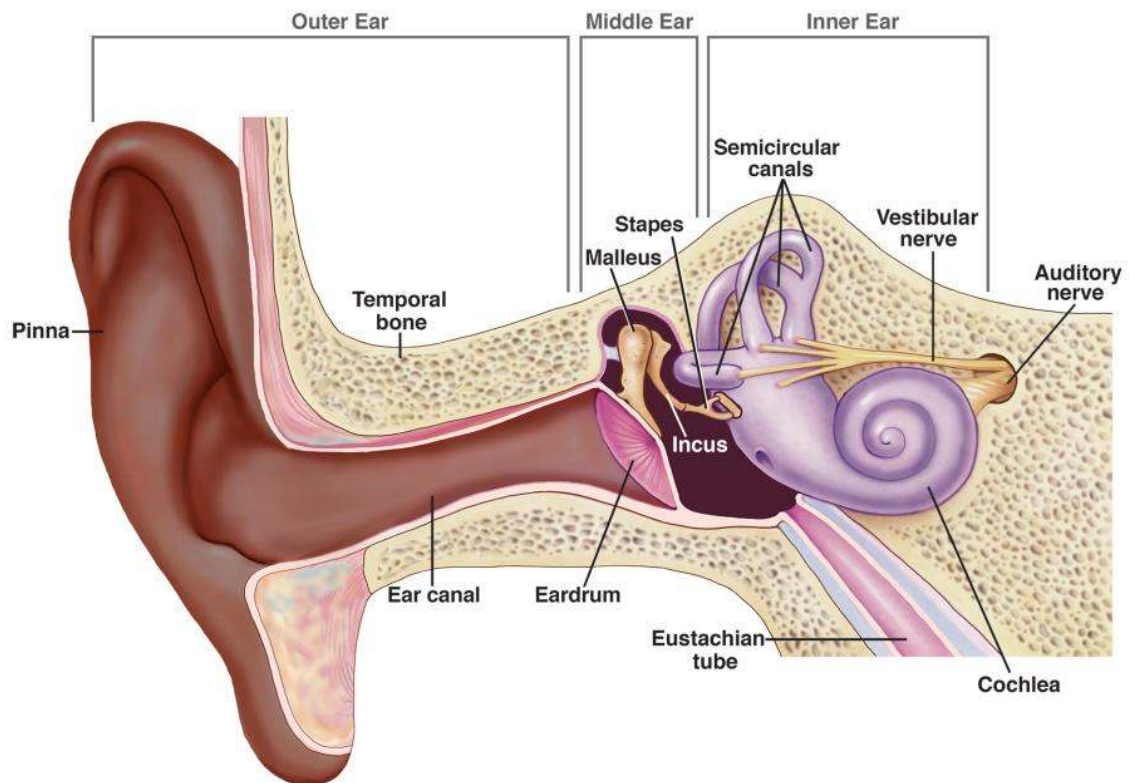
- Can be done by:
- Studying statistics on the incidence of disease

Developing Strategies to Prevent a Non-infectious Disease Outbreak:

- Type of strategies used dependent on characteristics of specific disease
- Normally has education programs + campaigns
- Distributed over many forms of media
- To:
 - raise awareness of disease
 - it's characteristics
 - effects on individual
 - suggestions to prevent development of disease
 - Suggestions about ways to lessen effects of disease
 - Support service advice
- Legislation:
 - Reduce exposure of population to risk factors of disease

Technologies and Disorders:

The Ear:



-

Structure and Function of the Ear:

- Sound waves (sound energy) → outer ear canal by pinna (external part of ear: skin + cartilage)
 - Create vibrations (kinetic energy)
 - Of identical frequency in the eardrum
- Vibrations → small bones in middle ear (ossicles)
 - Ossicles:
 - Hammer (malleus)
 - Anvil (incus)
 - Stirrup (stapes)
- Ossicles amplify (increase pressure of) vibrations + transfer kinetic energy → oval window
 - Ossicles amplify vibrations as:
 - Since eardrum area is much larger than base of stapes bone
 - The amount of vibrations on eardrum → stapes
 - Gets concentrated into smaller base area of stapes
 - Results in higher pressure vibrations
- Oval window:

- membrane that separates middle ear from inner ear
- Vibrations go through here from middle → inner ear
- Inner ear filled with fluid
- Vibrations → oval window → cochlea → round window
- Round window:
 - Acts as a valve
 - Relieves built up pressure from sound vibration waves
 - Set up in the perilymph (liquid) of the cochlea
- Cochlea:
 - Has receptor hair cells in Organ of Corti (basilar membrane)
 - Hair cells:
 - Convert sound vibrations in cochlea liquid → electrical impulses
 - Impulses sent through auditory nerve → brain

▶ Human ear - structure & working | Sound | Physics | Khan Academy

Hearing Loss:

Conductive Hearing Loss:

- Issue with mechanical movement of vibrations
 - Through outer + middle ear
- Causes:
 - Malformation of structures in middle/outer ear
 - Perforated eardrum
 - Infections in inner/middle ear
 - Damage to ossicles
 - Exposure to sudden loud noises (explosions, head trauma)
 - Hardening of stapes bone
- Normally affects loudness of sound

Sensorineural Hearing Loss:

- Affects:
 - Parts of cochlea
 - Hair cells
 - Auditory nerve
- Damage + hearing loss normally permanent
- Hearing loss affects:

- Loudness
- Clarity of sound
- Causes:
 - Excessive noise exposure (most common)
 - Heredity
 - Birth defects
 - Infections
 - Tumours
 - Medication
 - Aging
- Damage to inner ear + receptor hair cells:
 - Prevents kinetic energy of vibrations
 - Being turned into electrical impulses to be sent to brain

Technologies for Hearing Loss:

Conductive Hearing Loss:

Hearing Aids:

- Battery operated electronic device fitting into hollow of external ear
- Used to magnify sound vibrations
- Allow for better transmission from middle → inner ear
- Advantages:
 - Relatively cheap
 - Easy to install (no surgery)
- Disadvantages:
 - Amplify all sounds in environment
 - May cause pain
 - Not useful for severe damage to inner ear or auditory nerve

Bone Conduction Implant:

- Microphone that detects sound
- Implant in bone above ear
- Microphone turns sound into vibrations
- Vibrations → implant
- Vibrations directed through bone to cochlea
- Then processed as normal
- Bypass malformed or damaged outer or middle ear
 - Transferring vibrations directly to inner ear

- Inner ear must be working

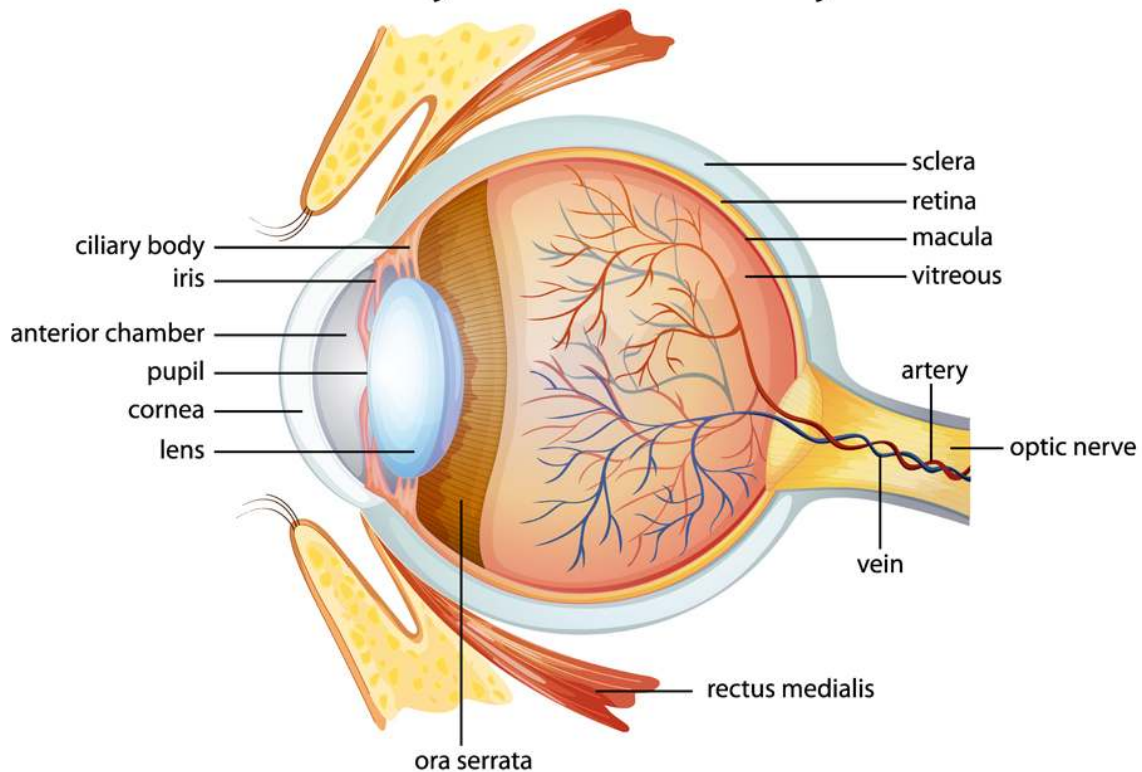
Sensorineural Hearing Loss:

Cochlear Implant:

- External microphone + speech processor
- Transmitter (on scalp)
- Receiver (implant under skin opposite the transmitter)
- mic + speech processor captures sound
- Processors sends digital signals to transmitter
- Receiver convert signals → electrical impulses
- Receiver sends electrical impulses to
 - electrode array surgically put inside cochlea
- Electrodes collect impulses
 - Send them to auditory nerve
- Impulses from auditory nerve → brain for processing
- Tries to mimic function of hair cells within cochlea
- Advantages:
 - Provides hearing to profoundly deaf people
- Disadvantages:
 - Expensive (surgery + ongoing costs)
 - Surgery side effects
 - infection, drooping face, numb tongue from nerve damage
 - Patient must learn to interpret sounds again
 - Limited scope if auditory nerve or brain region for processing damaged

The Eye:

Anatomy of the Human Eye



Structure and Function of the Eye:

- Sclera:
 - Tough protective coat forming majority of the eye
 - Transparent curved front section forms cornea
- Choroid:
 - Has most of blood vessels
 - Back of layer is black
 - To make eye "light tight"
 - Reduce scattering or reflection of light in eye
 - Front section forms:
 - Ciliary body
 - Lens
 - Iris (coloured part)
- Inner layer:
 - Retina:
 - Has light receptor cells:
 - Covers back of eye
- For clear, focused images to be seen:

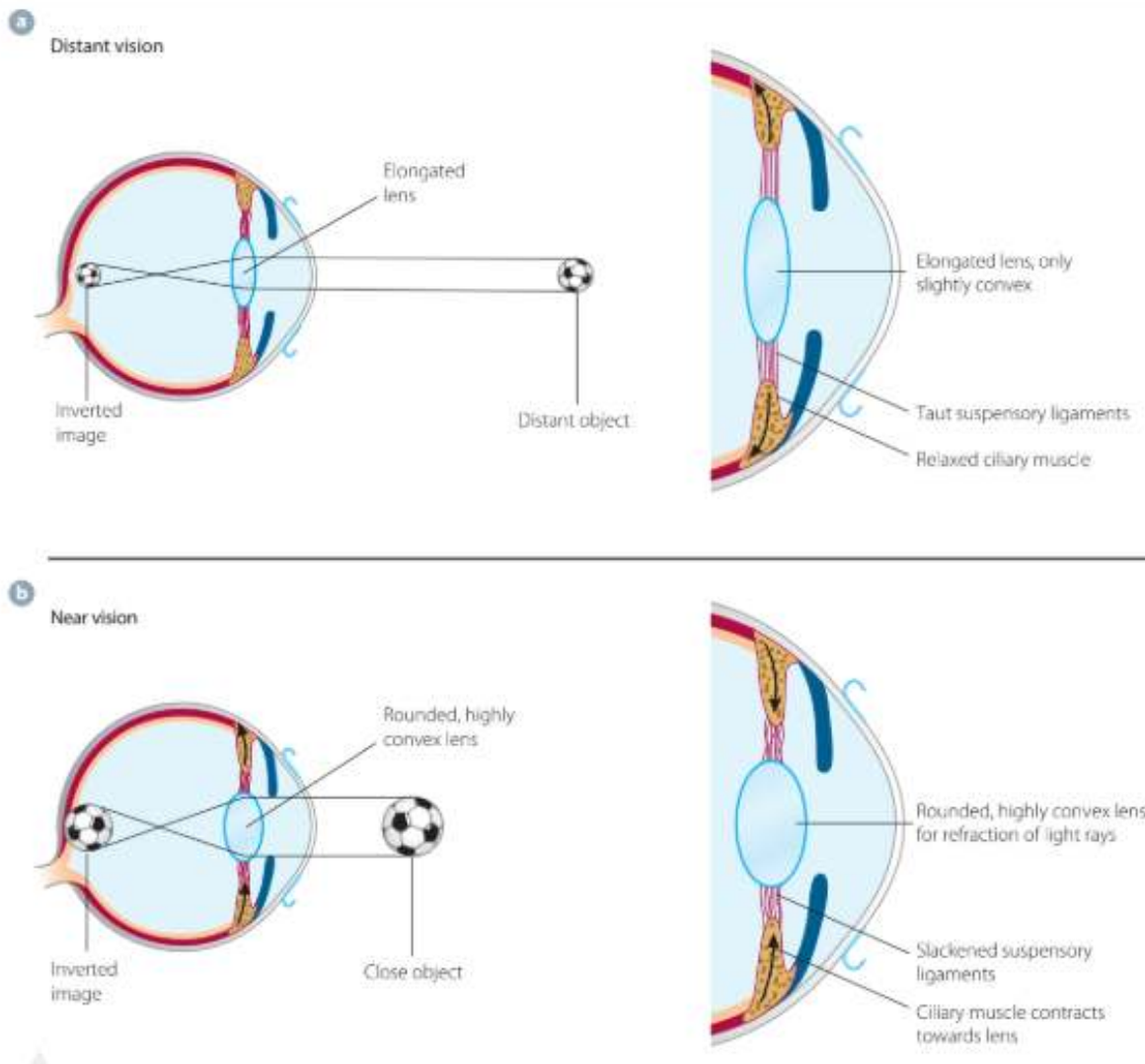
- Light must be refracted (bent)
 - To make sharp focus on retina
- Correct degree of refraction when
 - When all parts of eye are working properly
- Light reflected from object passes through cornea
 - Curve of cornea refracts incoming light
- Light → cornea → aqueous humour → pupil
- Aqueous humour:
 - Liquid
 - Between cornea, lens
- Pupil:
 - Black
 - Hole that can change size
 - By muscle iris
- Iris:
 - Coloured part of eye
 - Muscle
 - High light intensity:
 - Iris expands
 - Makes pupil smaller
 - Restricts amount of light entering eye
 - Low light intensity:
 - Iris contracts
 - Makes pupil larger
 - Increases amount of light entering eye
- Light → pupil → lens
- Lens:
 - Is Convex
 - Focuses light onto one point on retina surface
 - curve of lens
 - Changed by ciliary muscle
 - So lens refract incoming light
 - To make focused image on retina
 - Allows focused vision at different distances
 - Process of lens change curve
 - Accommodation
- Light → lens → vitreous humor

- Vitreous humour:
 - Clear jelly fluid
 - Maintains shape of eye
- Retina:
 - Has layers of nerve cells
- Fovea:
 - Central most sensitive part
 - Is a depression in the macula
 - Directly opposite incoming light
 - Has densely packed cone cells
 - No rod cells
- 2 types of photoreceptor cells:
 - Rods:
 - Perceive shades of grey
 - For Vision in low light
 - Cones:
 - Perceive red green blue
 - For Colour vision
 - Sharp central vision
 - closest to the back of the eye
 - Stimulated by light focused on retina
 - Turn light energy → electrical nerve impulses
 - Impulses → optic nerve → brain
 - Perceives images as upside down
 - When interpreted by brain it is flipped to rightside up
- Light → cornea → aqueous humour → pupil → lens → vitreous humour → retina → turn to electrical nerve impulses → brain

Accommodation:

- Lens change curve
 - To refract light by correct amount
 - To form sharply focused image on retina
 - For objects at any distance
- Light from far objects
 - Refracted less
 - As length for light to reach retina is already more
- Light from near objects
 - Refracted more

- As length for light to reach retina is less
 - Needing to be bent more to reach retina
- More curvature of lens
 - More light passing through is refracted
 - (greater refractive power)
- Lens curve changed by
 - Actions of ciliary muscles
 - Suspensory ligaments
- When viewing distant objects:
 - Ciliary muscles relax
 - Draws sclera back
 - Holds suspensory ligaments stretched
 - Lens is lengthened
 - Decrease curvature
- When suspensory muscle relaxes
 - It goes towards inside of eye
- Viewing close objects:
 - Ciliary muscles contract
 - Draws sclera forward
 - Release tension on suspensory ligaments
 - Lens is rounder
 - Increase curvature



Photoreceptor Cells - Rods and Cones:

- In retina
- Receive light energy
- Convert into electrochemical impulses
- Impulses sent to brain for interpretation as images
- Rods:
 - Perceive shades of grey
 - For Vision in low light
- Cones:
 - Perceive red green blue
 - For Colour vision
 - Sharp central vision

Visual Disorders:

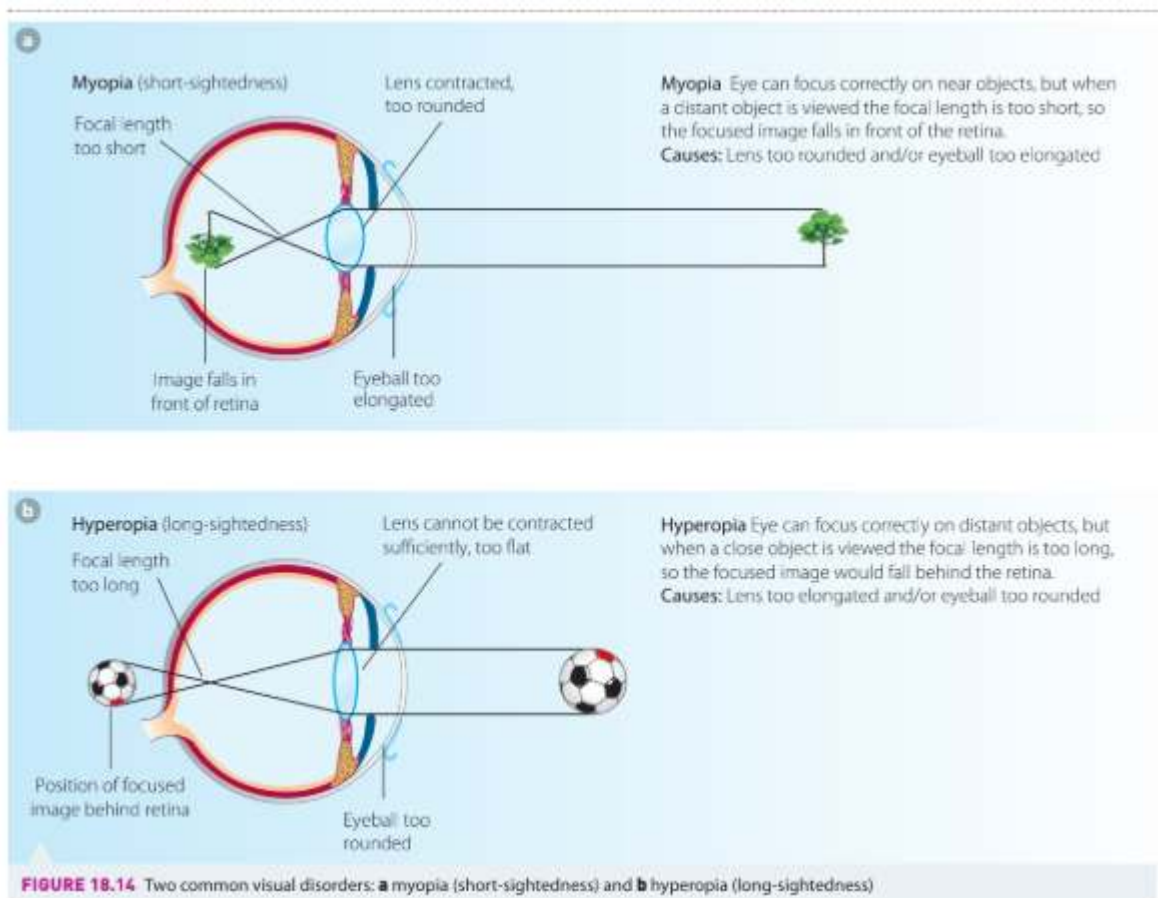
Long and Short Sightedness:

Short Sightedness (Myopia):

- See near objects
- Far objects are blurred
- Occurs because
 - Focused image from distant object
 - Is in front of retina
 - While image from near object
 - Is on retina
- Causes:
 - Shape of eye too elongated oval
 - Refractive power of cornea not enough for eye shape
 - Lens not flat enough when ciliary muscles contract

Long Sightedness (Hyperopia):

- See far objects clearly
- Near objects blurred
- Occurs because:
 - Focused image from near object
 - Is behind retina
 - While image from far object
 - Is on retina
- Causes:
 - Eye is short (too rounded)
 - Lens too flat
 - Can't achieve needed curveness
 - Due to loss of elasticity in old age
 - Refractive power too much for eye shape



- (Cause labels shown are separate, not impacted by one another)

Cataracts:

- Clouding of the lens
- Reduce transmission of light through lens
- Results in:
 - Blurred vision of
 - Near and far objects
 - Increased sensitivity to glare of bright sunlight

Retinopathy:

- Damage to retina
- Caused by
 - Changes to blood vessels surrounding the region

Macular Degeneration:

- Distortion or absence of fovea
- Degeneration of cells under retina
- Central vision associated with macula is affected
- Peripheral vision not affected

- Affects ability of:
 - Reading
 - Recognising faces
 - Driving
 - Everyday tasks
- Degeneration of these cells
 - Prevents light from being focused onto this area
 - Leading to loss of vision

Colour Blindness:

- One of the 3 types of cone cells
 - Not there
 - Not function properly
- Normally inherited
- Affects both eyes

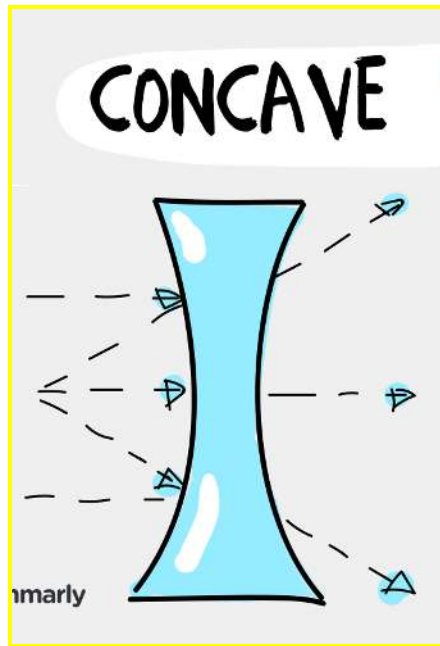
Technologies for Visual Disorders:

Spectacles (Glasses):

- Frames that hold clear lenses
 - made of clear hard plastic

- Myopia (short sightedness):

- Concave lens
 - Thick edges
 - Thin centre
 - Bend rays outwards
 - Diverge (spread) them before reaching eye
 - Extends focal length of light rays
 - Image on retina > in front of retina



- Concave → caving in

- Hyperopia (long):

- Convex lens:

- Thin edges

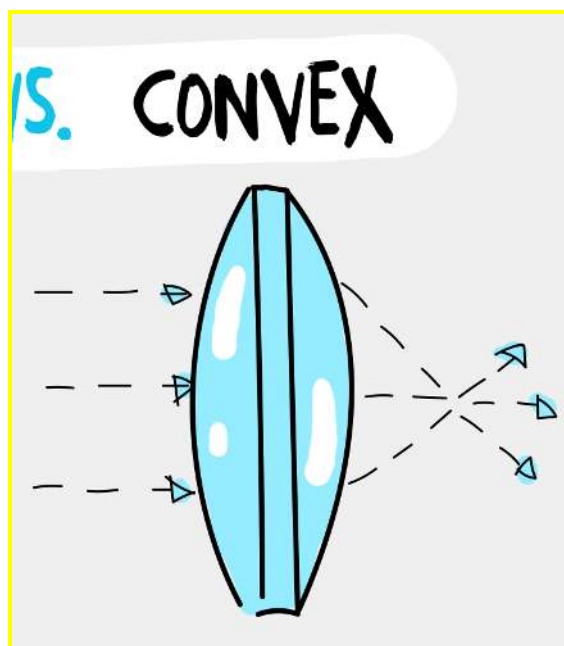
- Thick centre

- Bends light inwards

- Converge (merge) them before reaching eye

- Shorten focal length

- Image on retina > behind retina



- Where lenses are thicker are where light is going through them

- Concave → thicker edges, light going out to thicker edges

- Convex → thicker centre, light going into centre

Laser Eye Surgery:

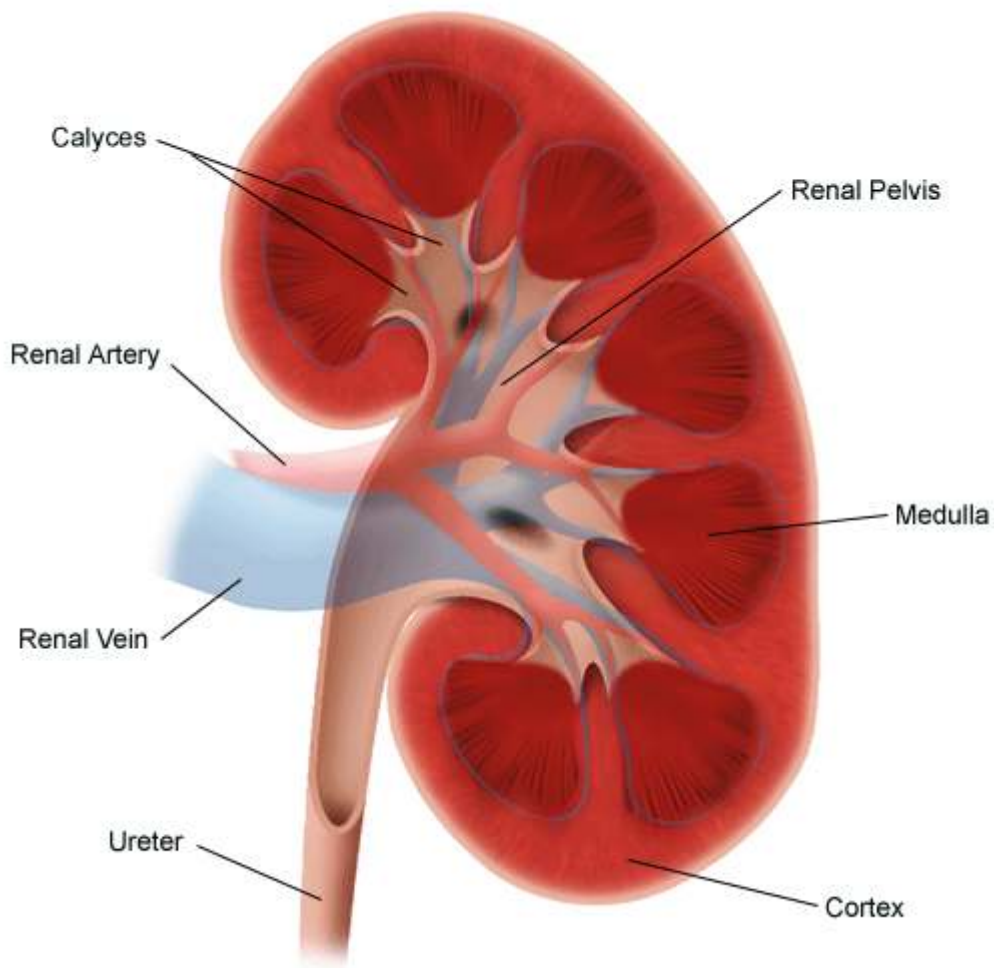
- Cut made in cornea
- Laser put onto cornea
 - changes curvature of cornea
- Allows light entering eye
 - To be refracted correctly
 - Forming focused image on retina

Cataract Surgery:

- Intraocular lens implantation
- Cloudy lens removed from lens capsule
- Artificial lens inserted
- Small incision in eye
 - Device inserted emitting high frequency sound
 - Breaks up cloudy lens
- Lens particles suctioned out
- Artificial lens inserted into lens capsule

The Kidney:

Anatomy of the Kidney



- Removes nitrogenous wastes from bloodstream
- Maintain optimal levels of salt, water (osmoregulation)
 - Homeostasis of these

Structure and Function of Kidney:

- 3 main functions:
 - Filtration
 - Reabsorption
 - Secretion
- 3 main regions:
 - Cortex
 - Medulla
 - Pelvis
- Nephron:
 - Functional unit of kidney
 - Extends from cortex → medulla

- Processes waste products from blood

- To create urine

Filtration:

- Blood from heart → aorta → renal artery → kidney

- Renal artery → blood vessels → capillaries

- Capillaries make glomerulus group

- Surrounded by bowman's capsule

- glomerulus + blood in glomerulus enters nephron

- Filtration based on size

- Substances that are small enough to be filtered are filtered

- Substances are:

- Nitrogenous waste products

- Large volumes of water having

- Amino acids

- Glucose

- Salts

- Substances then:

- Diffuse from capillaries → bowman's capsule (start of nephron)

- Aka glomerular filtrate

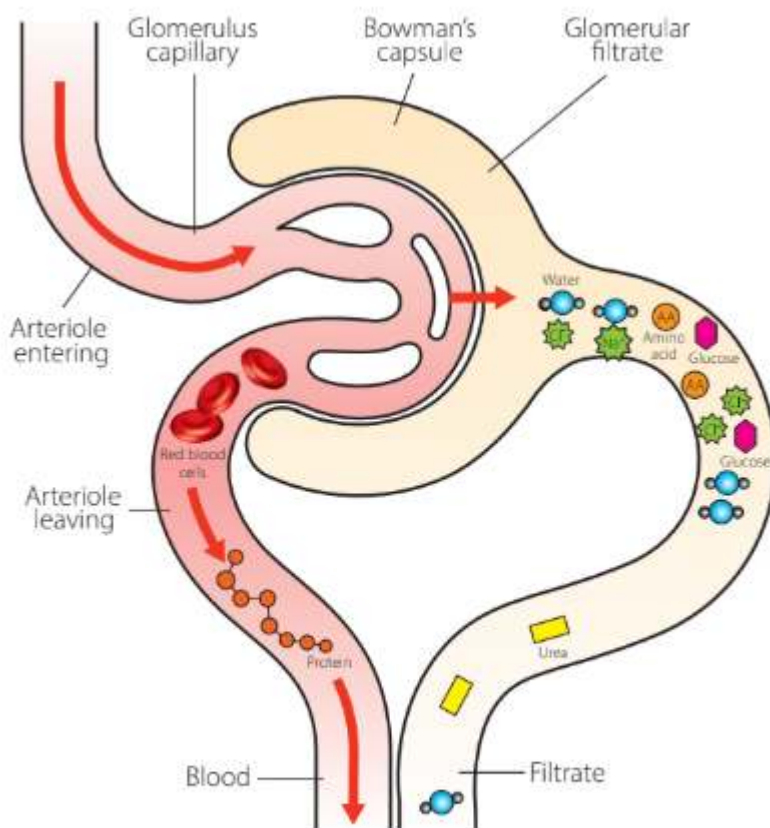


FIGURE 18.20 Filtration of blood in the Bowman's capsule

Reabsorption:

- Returns essential components filtered from blood → back to bloodstream
- Reabsorbed into blood:
 - Amino acids
 - Glucose
 - Ions
 - Calcium
 - Hydrogen bicarbonate
 - Vitamins
- All solutes reabsorbed from nephron
 - Move by
 - Active transport
 - Facilitated diffusion
 - Into proximal, distal tubules
- As solutes reabsorbed, water follows by osmosis
- Ascending (going up) loop of henle + collecting duct:
 - Many ions
 - Actively pumped into medulla tissues

- Causes conc. Gradient in tissues lower than in tubules
 - ∴ water moves
 - From descending (going down) loop of henle + collecting duct → tissues
- Reabsorp of water in all parts of tubules + collecting duct
 - Not in ascending loop of henle

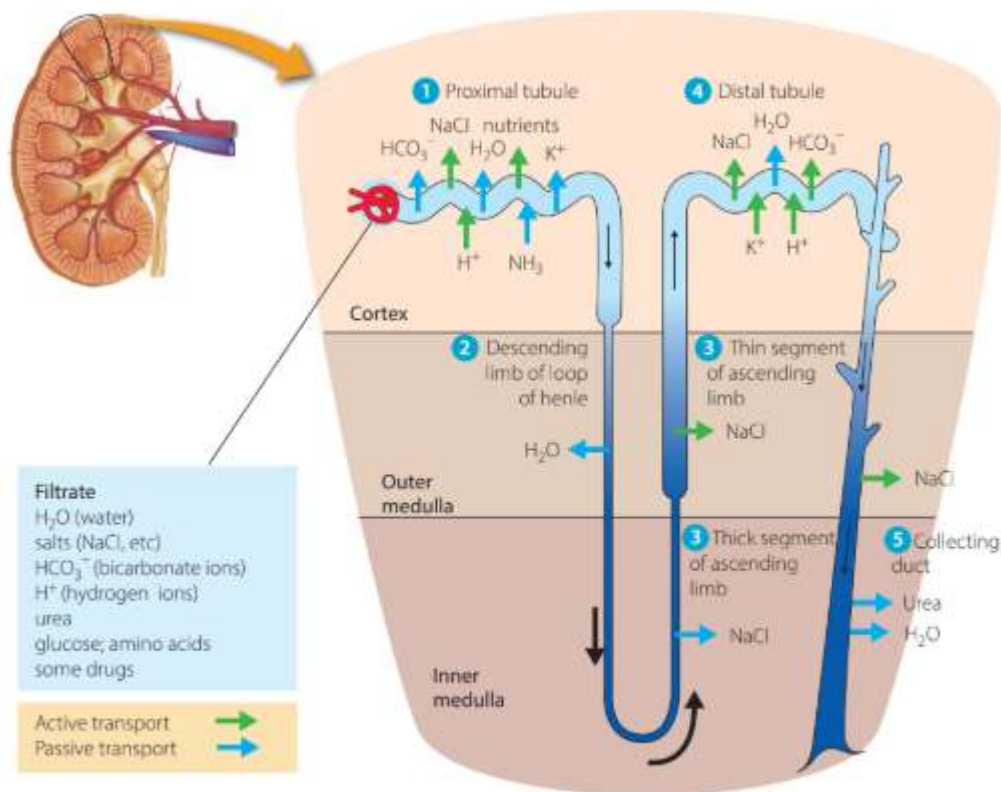


FIGURE 18.22 Reabsorption of ions and water from the tubules by active transport (ions) and passive transport (water). Blood vessels are not shown.

Secretion:

- **Removal of toxic substances**
 - **From blood capillaries + tissues surrounding tubules**
- Active movement of toxic substances → tubules
 - For removal
- Includes:
 - **Metabolic waste**
 - **Urea**
 - **Ammonia**
 - Hydrogen ions
 - Drugs
- Urea + ammonia
 - Moved by diffusion

- Everything else
 - Active transport
- H ions, saccharin, drugs → proximal nephron
- Urea → descending part of loop of henle

Hormonal Regulation of Osmoregulation:

- Hormone aldosterone
 - Stimulates reabsorption of salt in loop of henle
 - Regulate salt + water balance in kidney
- Hormone ADH (antidiuretic hormone):
 - Stimulates reabsorption of water in kidney

Removal:

- Water + nitrogenous wastes + other wastes
 - In collecting duct
 - is urine
- Urine → pelvis of kidney → ureters → bladder → urethra for elimination

Loss of Kidney Function:

- Occur by:
 - Damage to nephrons
 - Cant filter, reabsorb, secrete properly
- Results in failure to remove wastes
 - Not able to balance water, salt levels in blood

Kidney Disorders:

- Result in:
 - Build up of metabolic waste in blood
 - Imbalanced amounts of salt + water in body fluids

Diabetic Nephropathy:

- Kidney damage caused by diabetes

Kidney Stones:

- Form when waste products crystallise into lumps
- Painful
- Cause blockages
- Factors increasing risk:
 - Dehydration

- Poor diet
- Hereditary factors

Causes of Loss of Function:

- Type 1, 2 diabetes
 - High levels of blood glucose
 - Kidneys filter more blood
 - More stress on nephrons
 - Damage to nephrons
 - Larger molecules leak into nephrons
 - Cause blockages, more damage
- Continuous high bp:
 - Constant high force of blood
 - Pushed through walls of blood vessels of glomerulus → bowman's capsule
 - Damages nephrons
 - Large molecules leak into tubules
- Recurrent kidney infections:
 - Damage to nephrons
 - + prolonged use of drugs, medicine to treat infections
 - Reduces kidney function
- Kidney stones, tumours, anything blocking passage of urine
 - Urine goes back into kidney
 - Builds up pressure
 - → kidney damage
 - → complete shutdown

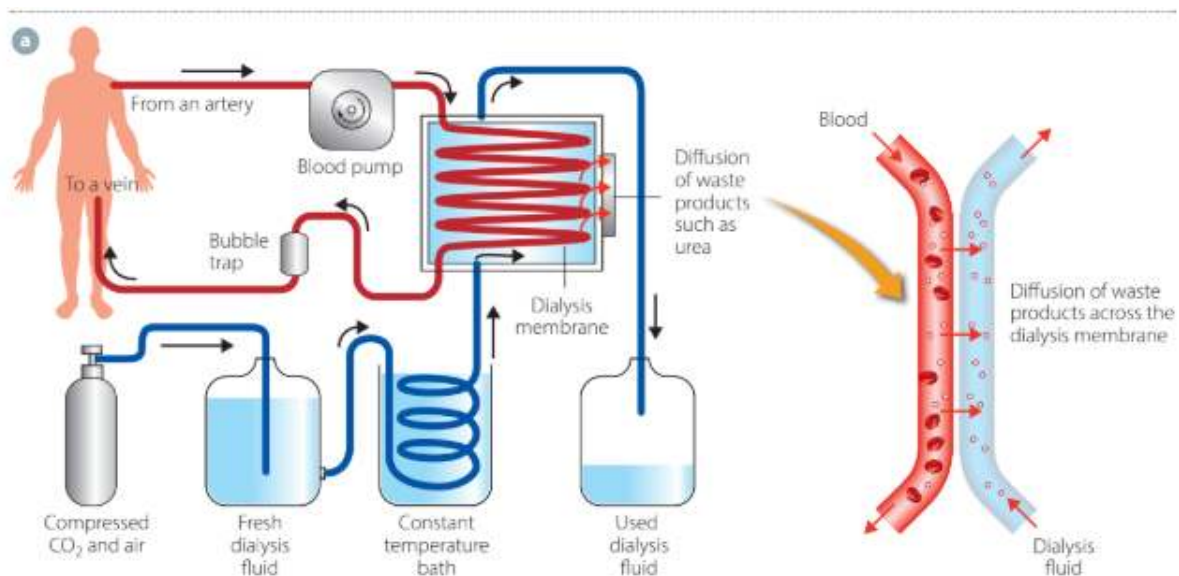
Technology for Loss of Kidney Function:

- 2 types of renal dialysis
 - Carries out some functions of kidneys

Haemodialysis:

- Function of dialysis:
 - Remove metabolic wastes
 - Built up in blood
- Patient connected to dialysis machine
- Pumps blood through system of tubes
 - Tubes:
 - Are coiled

- Increase surface area
- $\therefore$ increase rate of diffusion
- Have artificial semipermeable membranes
- Tubes submerged in dialysis fluid (dialysate)
 - Flows in opposite direction to blood
 - Maintains concentration gradient
 - Maximise diffusion
- Dialysate
 - Same conc. as blood plasma
 - Without metabolic wastes
- Conc. of metabolic waste higher in blood than in dialysate
 - Waste moves from semi permeable membrane $\rightarrow$ dialysate
 - Using diffusion (high conc to low conc)
- Dialysate continually replaced
- Renal dialysis:
 - Done for 3 - 4 hours
 - 2 - 3 times a week



Advantages:	Disadvantages:
- Good for individuals with kidney failure waiting for kidney transplant - Allows for removal of - Toxins - Excess fluid	- Time consuming - Limited amounts of fluid, wastes removed from blood - Substances like - Sodium phosphate

From blood - Of individuals with kidney failure	- Potassium ions - Don't diffuse quickly enough - Accumulate in blood
--	---

Peritoneal Dialysis:

- Dialysate fluid put into abdominal cavity
 - Through abdominal lining
 - Peritoneum
- Dialysate fluid absorbs waste products + extra blood fluid
- Dialysate then drained out
 - Takes waste with it out of body

Effectiveness of Technologies:

- "Evaluate" meaning in dot point:
 - List components
 - Describe characteristics and features of
 - Explain purpose and function
 - Provide examples
 - Determine effects and impacts
 - Make a judgement
 - Determine effectiveness of technology

Cochlear Implant:

- Before technology:
 - Individuals cope themselves with disorder
 - Affected:
 - Individuals
 - Family
 - Carers
 - Burden on health, welfare systems
 - Increased because of
 - Continuous support to be provided
- For people with sensorineural hearing loss
- Highly effective technology
- Gives individuals ability to communicate effectively with those around them
 - Before having:
 - to communicate in other ways

- Sign language
 - Lip reading
- Reduced ability to develop language skills
 - Especially in those born profoundly deaf
- Leading to:
 - reduced quality of life
 - Limited ability to function + express themselves
 - At higher levels in community
- Those born profoundly deaf can now
 - Develop at normal rate
 - Show no signs of developmental delays in
 - Language
 - Hearing
 - Speech
 - Integrate into society
- Can be fitted into children as young as 6 months
 - Developing language + speaking skills normally
 - No need to attend special schools
- Older adults:
 - After adjustment, training
 - Can hear world again
 - Experience improved quality of life
 - Can be more independent
 - Contribute positively to family, society
 - Without extra assistance
 - Otherwise adding
 - Extra bother
 - Extra costs
- Disadvantages:
 - Undergo operation surgery for implanting device
 - Postoperative side effects
 - Ongoing costs for updating
 - Programming, training
 - To teach person to interpret hearing of sounds
 - Limited distance of hearing
- Advantages of:
 - Hearing, developing speech normally

- Being highly functioning, contributing member of society with no disability
- Far outweighs disadvantages